

$$\begin{aligned}
& \left(-\frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_d^P] \delta_{i1i2} + \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_d^P] \delta_{i1i2} - \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_d^P] \delta_{i1i2} - \right. \\
& \frac{1}{9} \sum_p g_1^4 LF_{3,0}[m_e^P] \delta_{i1i2} + \frac{5}{24} \sum_p g_1^4 LF_{4,-1}[m_e^P] \delta_{i1i2} - \frac{4}{45} \sum_p g_1^4 LF_{5,-2}[m_e^P] \delta_{i1i2} - \\
& \frac{1}{18} \sum_p g_1^4 LF_{3,0}[m_l^P] \delta_{i1i2} + \frac{5}{48} \sum_p g_1^4 LF_{4,-1}[m_l^P] \delta_{i1i2} - \frac{2}{45} \sum_p g_1^4 LF_{5,-2}[m_l^P] \delta_{i1i2} - \\
& \frac{1}{54} \sum_p g_1^4 LF_{3,0}[m_q^P] \delta_{i1i2} + \frac{5}{144} \sum_p g_1^4 LF_{4,-1}[m_q^P] \delta_{i1i2} - \frac{2}{135} \sum_p g_1^4 LF_{5,-2}[m_q^P] \delta_{i1i2} - \\
& \frac{4}{27} \sum_p g_1^4 LF_{3,0}[m_u^P] \delta_{i1i2} + \frac{5}{18} \sum_p g_1^4 LF_{4,-1}[m_u^P] \delta_{i1i2} - \frac{16}{135} \sum_p g_1^4 LF_{5,-2}[m_u^P] \delta_{i1i2} - \\
& \frac{1}{18} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} - \frac{1}{12} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} + \frac{4}{45} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{96} g_1^4 LF_{2,1,0}[m_1, m_l^{-i1}] \delta_{i1i2} - \frac{1}{96} g_1^4 LF_{2,2,-1}[m_1, m_l^{-i1}] \delta_{i1i2} + \frac{1}{48} g_1^4 LF_{3,1,-1}[m_1, m_l^{-i1}] \delta_{i1i2} - \\
& \frac{1}{96} g_1^4 LF_{4,1,-2}[m_1, m_l^{-i1}] \delta_{i1i2} - \frac{1}{96} g_1^4 LF_{2,1,0}[m_1, m_l^{-i2}] \delta_{i1i2} - \frac{1}{96} g_1^4 LF_{2,2,-1}[m_1, m_l^{-i2}] \delta_{i1i2} - \\
& \frac{1}{48} g_1^4 LF_{3,1,-1}[m_1, m_l^{-i2}] \delta_{i1i2} - \frac{1}{96} g_1^4 LF_{4,1,-2}[m_1, m_l^{-i2}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_l^{-i1}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_l^{-i1}] \delta_{i1i2} + \\
& \frac{1}{16} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_l^{-i1}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_l^{-i1}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_l^{-i2}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_l^{-i2}] \delta_{i1i2} + \\
& \frac{1}{16} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_l^{-i2}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_l^{-i2}] \delta_{i1i2} - \\
& \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{2,2,0}[m_d^r, m_q^P] \delta_{i1i2} + \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{3,2,-1}[m_d^r, m_q^P] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{2,1,0}[m_e^P, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{2,2,-1}[m_e^P, \tilde{\mu}] - \\
& \frac{1}{12} g_1^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{3,1,-1}[m_e^P, \tilde{\mu}] - \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{2,2,0}[m_e^r, m_l^P] \delta_{i1i2} + \\
& \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{3,2,-1}[m_e^r, m_l^P] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{3,1,0}[m_l^P, m_e^r] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{3,2,-1}[m_l^P, m_e^r] \delta_{i1i2} - \frac{3}{8} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{4,1,-1}[m_l^P, m_e^r] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_e^{pr} y_e^{pr} LF_{5,1,-2}[m_l^P, m_e^r] \delta_{i1i2} + \frac{1}{48} g_1^4 LF_{2,1,0}[m_l^{-i1}, m_1] \delta_{i1i2} - \\
& \frac{1}{96} g_1^4 LF_{3,1,-1}[m_l^{-i1}, m_1] \delta_{i1i2} + \frac{1}{16} g_1^2 g_2^2 LF_{2,1,0}[m_l^{-i1}, m_2] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{-i1}, m_2] \delta_{i1i2} + \frac{1}{48} g_1^4 LF_{2,1,0}[m_l^{-i2}, m_1] \delta_{i1i2} - \\
& \frac{1}{32} g_1^4 LF_{3,1,-1}[m_l^{-i2}, m_1] \delta_{i1i2} + \frac{1}{16} g_1^2 g_2^2 LF_{2,1,0}[m_l^{-i2}, m_2] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{-i2}, m_2] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{3,1,0}[m_q^P, m_d^r] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{3,2,-1}[m_q^P, m_d^r] \delta_{i1i2} - \frac{5}{8} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{4,1,-1}[m_q^P, m_d^r] \delta_{i1i2} + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y}_d^{pr} y_d^{pr} LF_{5,1,-2}[m_q^P, m_d^r] \delta_{i1i2} + \frac{1}{12} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{2,2,0}[m_q^P, m_u^r] \delta_{i1i2} - \\
& \frac{1}{24} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{3,2,-1}[m_q^P, m_u^r] \delta_{i1i2} - \frac{1}{12} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{3,1,0}[m_u^r, m_q^P] \delta_{i1i2} - \\
& \frac{1}{12} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{3,2,-1}[m_u^r, m_q^P] \delta_{i1i2} - \frac{1}{4} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{4,1,-1}[m_u^r, m_q^P] \delta_{i1i2} + \\
& \frac{1}{2} g_1^2 \overline{a}_u^{pr} a_u^{pr} LF_{5,1,-2}[m_u^r, m_q^P] \delta_{i1i2} + \frac{5}{24} g_1^4 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} - \\
& \frac{1}{6} g_1^4 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{8} g_1^2 g_2^2 LF_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{2} g_1^2 g_2^2 LF_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} - \\
& \frac{1}{12} g_1^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{2,1,0}[\tilde{\mu}, m_e^P] + \frac{1}{24} g_1^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_e^P] - \\
& \frac{1}{24} g_1^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_e^P] - \frac{1}{32} g_1^4 LF_{2,1,1,-1}[m_1, m_l^{-i1}, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{16} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_l^{-i1}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{32} g_1^4 LF_{2,1,1,-1}[m_1, m_l^{-i2}, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{16} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_l^{-i2}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y}_e^{-i2p} y_e^{i1p} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^P] - \\
& \frac{1}{32} g_1^4 LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_l^{-i1}] \delta_{i1i2} + \frac{1}{32} g_1^4 m_1^2 LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{-i1}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^4 LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_l^{-i2}] \delta_{i1i2} + \frac{1}{32} g_1^4 m_1^2 LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{-i2}] \delta_{i1i2} - \\
& \frac{3}{32} g_2^4 LF_{2,1,1,-1}[m_2, m_l^{-i1}, \tilde{\mu}] \delta_{i1i2} - \frac{3}{16} g_2^4 m_2^2 LF_{2,1,1,0}[m_2, m_l^{-i1}, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{3}{32} g_2^4 LF_{2,1,1,-1}[m_2, m_l^{-i2}, \tilde{\mu}] \delta_{i1i2} - \frac{3}{16} g_2^4 m_2^2 LF_{2,1,1,0}[m_2, m_l^{-i2}, \tilde{\mu}] \delta_{i1i2} + \\
& \frac{3}{16} g_2^2 \tilde{\mu}^2 \overline{y}_e^{-i2p} y_e^{$$